

Question number	Answer	Mark
4(a)	B flexible, non-elastic tissue connecting muscle to bone flexible tissue connecting bone to bone	(1)

Question number	Answer	Mark
4(b)(i)	A P to T	(1)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An answer that includes three of the following points: similarities: • contain {actin / myosin / tropomyosin / troponin} (1) • consist of {sarcoplasm / sarcolemma / sarcomeres} (1) differences: • slow-twitch muscle fibres have more mitochondria (1) • slow-twitch muscle fibres have more myoglobin (1) • slow-twitch muscle fibres have less glycogen (1) • slow-twitch muscle fibres have less creatine phosphate (1)	To gain maximum marks, at least one similarity must be included in the answer Accept the converse points for fast-twitch fibres	(4)

Question number	Answer	Mark
4(c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content • power output is (always) greater in muscles containing more than 50% fast-twitch muscle fibres • power output in muscles containing more than 50% fast-twitch muscle fibres increases with speed of contraction. • power output in muscles containing less than 50% fast-twitch muscle fibres decreases with higher speeds of contraction. • fast-twitch muscles are designed for rapid short bursts of energy. • therefore power output will increase with speed of contraction. • because they are adapted for anaerobic respiration. • example of adaptation described • muscles with low content of fast-twitch muscles cannot sustain aerobic respiration for long periods of time	(6)

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information with some attempt made to link knowledge and understanding to the given context.
2	3-4	An explanation will be given with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning with some structure.
3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.

